MILITARY SPECIFICATION SHEET

# CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, UNARMORED M17/74-RG213, AND ARMORED M17/74-RG215

| THIS CABLE USES PVC MATERIAL AND IS NOT TO BE USED IN AEROSPACE APPLICATIONS. |

| NOTE: THE AIR FORCE HAS RESTRICTED THE USE OF PVC IN AEROSPACE AND GROUND SUPPORT APPLICATIONS. CABLES WITH PVC JACKETING SHALL BE USED FOR RETROFIT PURPOSES ONLY UNTIL AN ALTERNATE JACKET IS APPROVED. |

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-17.

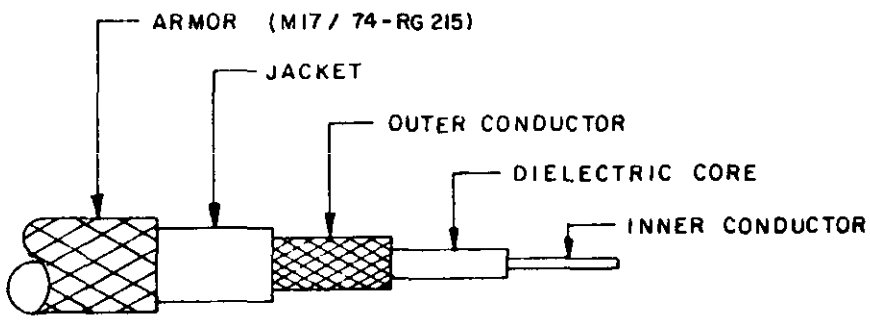



FIGURE 1. *Configuration*.



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TABLE I. Description.

| Component | Construction details |
|---|---|
| Inner conductor | Seven strands of bare copper wire, each strand 0.0296 inch diameter.<br>Overall diameter: 0.0888 inch ±0.0010. |
| Dielectric core | Type A-1: Solid polyethlyene.<br>Diameter: 0.285 inch ±0.007. |
| Outer conductor | Single braid of AWG No. 33 bare copper wire.<br>Diameter: 0.340 inch maximum.<br>Coverage : 95.3% nominal<br>Carriers : 24<br>Ends : 8<br>Picks/inch: 6.5 ±10% |
| Jacket | Type IIa: PVC.<br>Diameter: 0.405 inch ±0.007. |
| Armor (M17/74-RG215) | Single braid of aluminum-alloy wire.<br>Diameter: 0.475 inch maximum. |

ENGINEERING INFORMATION:

Continuous working voltage: 3,700 V rms, maximum.

Operating frequency: 1 GHz, maximum.

Velocity of propagation: 65.9 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -40°C to +85°C.

Inner conductor properties:

(C) DC resistance (maximum at 20°C): 0.176 ohm per 100 feet.

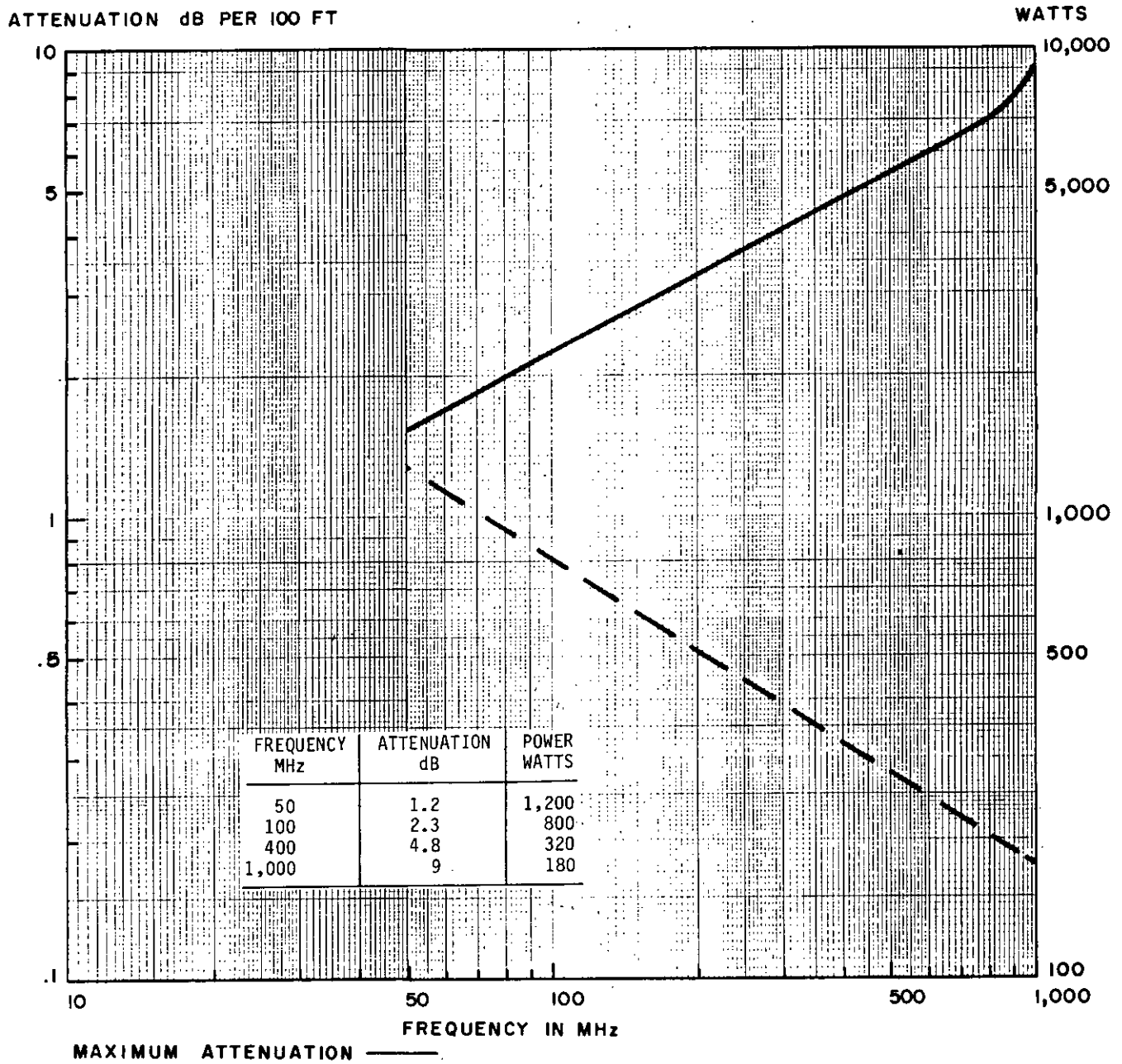
Elongation: 25 percent, minimum.

Tensile strength: Not applicable.

Engineering note: This cable is useful in general purpose, medium low temperature applications. (See connector series "N", "C", and "SC" per MIL-C-39012. For M17/74-RG213, NATO preferred type NWR-1; for M17/74-RG215, NATO preferred type NWR-17.)



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| FREQUENCY MHz | ATTENUATION dB | POWER WATTS |
|---|---|---|
| 50 | 1.2 | 1,200 |
| 100 | 2.3 | 800 |
| 400 | 4.8 | 320 |
| 1,000 | 9 | 180 |

Tabulated values are for reference only.
The values on the chart represent the requirements.

FIGURE 2. Power rating and attenuation.



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REQUIREMENTS:

Dimensions, configuration, and description: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination:

Out-of-roundness: Not applicable.

Eccentricity: 10 percent maximum.

Adhesion of conductors:

(C) Inner conductor to core: 9 pounds, minimum; 90 pounds, maximum.

Aging stability: +98°C ±2°C.

Stress crack resistance: Not applicable.

Outer conductor integrity: Not applicable.

(C) Cold bend: -40°C ±2°C.

Dimensional stability:

Inner conductor from core: 0.062 inch, maximum.

Inner conductor from jacket: 0.125 inch, maximum.

Contamination: Applicable.

Bendability: Not applicable.

Flammability: Not applicable.

(C) Weight:

M17/74-RG213: 0.111 pound per foot, maximum.
M17/74-RG215: 0.138 pound per foot, maximum.

Electrical:

Continuity: Applicable.

(C) Spark test: 5,000 V rms, +10%, -0%.

(C) Voltage withstanding: 10,000 V rms, +10%, -0%.

Insulation resistance: Not applicable.

Corona extinction voltage: 5,000 V rms, minimum.

Characteristic impedance: 50 ohms ±2.

Attenuation: See figure 2.

Structural return loss: See figure 3.

(C) Capacitance: 32.2 pF per foot, maximum.

Capacitance stability: Not applicable.

Capacitance unbalance: Not applicable.



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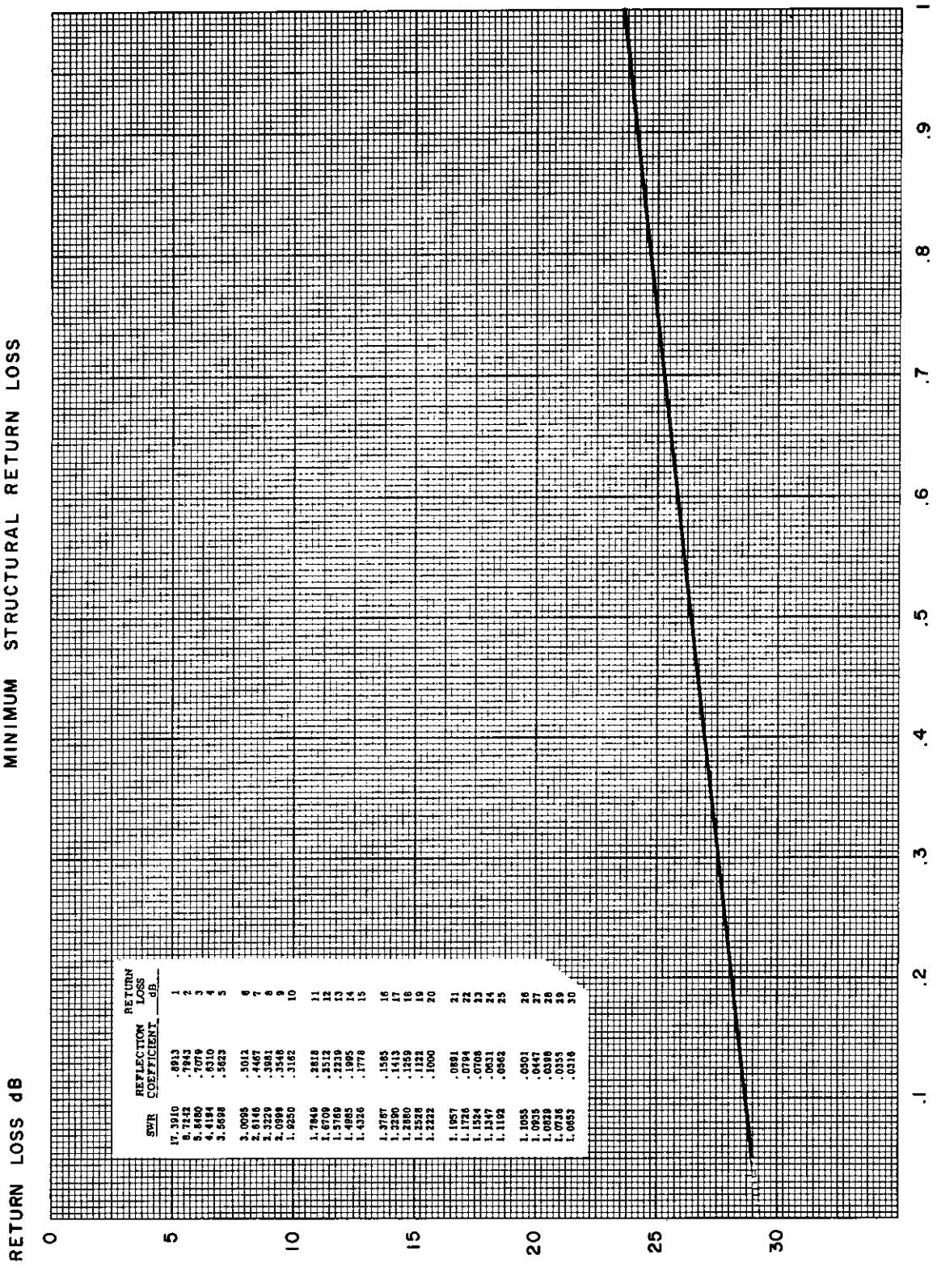



FIGURE 3. Structural return loss.



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Transmission unbalance: Not applicable.

Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

Part number: See table II.

Supersession data: See table II.

TABLE II. Cross-reference of part number.

| Part number | Superseded part number or type designation |
|---|---|
| M17/74-RG213 | |
| M17/74-RG215 | RG-215/U; RG-10A/U per MIL-C-17/5 (canceled) |

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH, TD
Air Force - 11, 17, 99
DLA - ES, IS

User activities:
Army - AR, AT, ME
Navy - AS, MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

(Project 6145-0911-13)



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